

Prajna G. Malettira



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EDUCATION

Purdue University, West Lafayette, Indiana, USA

Aug., 2022 - Present

Ph.D., Electrical and Computer Engineering

PES University, Bengaluru, Karnataka, India

Aug., 2018 - May, 2022

B.Tech., Electronics and Communication Engineering

SKILLS

Languages & Tools: Python · Bash · Git · MATLAB · Linux · Slurm **Frameworks:** PyTorch · TensorFlow · $\text{\LaTeX}$ · OpenCV

PUBLICATIONS

- **Malettira, P. G.**, Negi, S., Ponghiran, W., & Roy, K "TSkips: Efficiency through explicit temporal delay connections in spiking neural networks." **Accepted at TMLR 2025.** [\[Paper\]](#)
- **Malettira, P. G.**, Nagaraj, M., Roy, A., Negi, S., & Roy, K. "TraceNAS: Zero-shot LLM Pruning via Gradient Trace Correlation." **Under Review at NeurIPS 2026.** [\[Paper\]](#)
- Roy, A., **Malettira, P.G.**, Nagaraj, M. and Roy, K., 2025. Unifying Global Topology Manifolds and Local Persistent Homology for Data Pruning. **Accepted at NeurIPS 2025 Symmetry and Geometry in Neural Representations Workshop.** [\[Paper\]](#)
- Roy, A., **Malettira, P. G.**, Nagaraj, M., & Roy, K. (2026). TopoPrune: Robust Data Pruning via Unified Latent Space Topology." **Under Review at NeurIPS 2026.** [\[Paper\]](#)

RESEARCH EXPERIENCE

TraceNAS: Zero-shot LLM Pruning via Gradient Trace Correlation

Under Review

Python, Zero-shot NAS, LLM Pruning, Model Compression

- Developed a **scalable, training-free NAS framework for the joint compression of LLM depth and width**, discovering high performing non-uniform architectures across **LLaMA** and **Qwen** models at 2B to 70B parameter scales.
- Proposed a statistically grounded, scale-invariant and training-free proxy for compressed model performance. We achieve superior Spearman ($\rho = 0.94$) and Kendall Tau ($\tau = 0.82$) with average downstream accuracy.
- **Key takeaway:** Attained a **4.22 $\times$ prefill speedup**, reduced peak memory and maintains high-fidelity performance across reasoning tasks with **10 $\times$ faster compressed model discovery**.

TopoPrune: Robust Data Pruning via Unified Latent Space Topology.

Under Review; [NeurIPS Workshop](#)

Python, Topological Data Analysis, Persistent Homology

- Developed a **novel framework for topologically grounded dataset pruning**. The method resolves instability in traditional geometric data pruning methods, resulting in highly representative data subsets.
- Proposed the **first persistence homology method for data pruning** to determine local sample importance and optimize the corrective displacement of data points.
- **Key takeaway:** Improved the accuracy of models trained on coresets by up to **1.5%**, improved precision across random seeds by up to **6.5x** at high pruning rates.

Geometric Diffusion Pruning (TGIP): Curvature-Aware Compression of Generative Models

WIP

Python, Multimodal Diffusion Transformers, Flow Matching, Differential Geometry, Graph Neural Networks

- Developing a training-free pruning framework for multimodal Diffusion and Flow Matching models that leverages **Local Spectral Isometry** to identify and preserve the underlying manifold geometry of compressed models.
- **Key takeaway:** Initial results showcase FID and CLIP scores comparable with Euclidean-based metrics across pruning ratios, outperforming magnitude-based baselines by prioritizing topological consistency.

Exploring Model Memorization and Data Leakage via Sparse Auto Encoders

WIP

Python, SAEs, Interpretability, Dictionary Learning, Computer Vision

- Designing an analysis framework using **Sparse Auto Encoders (SAEs) and influence functions** to create an efficient memorization proxy for data sample memorization that is independent of large model training.
- **Key takeaway:** Preliminary results showcase correlation of $\rho = 0.5$ with established memorization baselines.

RELEVANT COURSEWORK

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|-------------------------|-----------------------|----------------------|-----------------------|
| • Computational Imaging | • Computer Vision | • Deep Learning (DL) | • Linear Algebra |
| • Generative Models | • Optimization for DL | • Random Variables | • Pattern Recognition |